

NER (Named Entity Recognition) Tags across Packages

BERT, spaCy, Stanford CoreNLP, Stanza

Contents

NER Approaches in the NLP Suite.....	1
The	1
BERT	1
spaCy.....	1
Stanford CoreNLP	2
Stanza.....	3

NER Approaches in the NLP Suite

The

BERT

The model currently available in the NLP Suite for NER with BERT only processes the English language.

geo for geographical entity

org for organization entity

per for person entity

gpe for geopolitical entity

tim for time indicator entity

art for artifact entity

eve for event entity

nat for natural phenomenon entity

O is assigned if a word doesn't belong to any entity.

spaCy

spaCy includes a list of 18 entity types:

Entity Type	Type Description
CARDINAL	A plain number that isn't another type - e.g., 'three', '1,000'.
DATE	A date or time period - e.g., 'June 2026', 'last summer'.
EVENT	A named event - wars, battles, hurricanes, sports events (e.g., 'World War II').

Entity Type	Type Description
FAC	FAC = 'facility'. A man-made structure - buildings, airports, highways, bridges.
GPE	GPE = 'geo-political entity'. A place that has a government - a country, a state/region, or a city (e.g., 'Italy', 'Atlanta').
LANGUAGE	A named language - e.g., 'English', 'Italian'.
LAW	A named law or legal document - e.g., 'the Constitution'.
LOC	A natural place that is NOT a country or city - mountains, rivers, lakes, oceans, regions (e.g., 'the Alps', 'the Nile').
MONEY	An amount of money - e.g., '\$5', 'ten euros'.
NORP	NORP = Nationalities, Religious, or Political groups - e.g., 'Italians', 'Catholics', 'Democrats'.
ORDINAL	A position word - 'first', 'second', 'third'.
ORG	An organization - a company, agency, school, or team (e.g., 'Emory University').
PERCENT	A percentage - e.g., '20%'.
PERSON	A person, real or fictional - e.g., 'Harry Potter'.
PRODUCT	An object or thing - vehicles, foods, gadgets (not services).
QUANTITY	A measurement - weight or distance (e.g., '5 km', 'two pounds').
TIME	A time shorter than a day - e.g., '3 p.m.', 'two hours'.
WORK_OF_ART	The title of a creative work - a book, song, or painting.

Stanford CoreNLP

PERSON
 ORGANIZATION
 MISC
 MONEY
 NUMBER
 ORDINAL
 PERCENT
 DATE
 TIME
 DURATION
 SETE
 MAIL
 URL
 CITY
 STATE_OR_PROVINCE
 COUNTRY
 LOCATION

NATIONALITY
RELIGION
TITLE
IDEOLOGY
CRIMINAL_CHARGE
CAUSE_OF_DEATH

Stanza

https://stanfordnlp.github.io/stanza/ner_models.html

Stanza's English model uses the same 18 OntoNotes entity types as spaCy - see the spaCy table above for a plain-language description of each (PERSON, GPE, LOC, FAC, NORP, ORG, DATE, and so on).

Important: Stanza tags cities, states, and countries all as GPE. Unlike Stanford CoreNLP, it does NOT separate CITY, STATE_OR_PROVINCE, and COUNTRY.

Stanza's other (non-English) languages use a smaller set of tags. The most common ones mean:

PER / PERS / PERSON - a person.

LOC / LOCATION - a place (city, country, region, or natural feature). In languages such as Italian, French, German and Spanish, every place is tagged LOC.

ORG / ORGANIZATION - an organization (company, agency, institution).

MISC / MISCELLANEOUS - a named thing that does not fit the other groups.

EVT / EVENT - a named event. PRO / PRODUCT - a product or object. DATE / TIME - a date / a time.

NE, NUM, PNAME, RACE (Burmese) - named entity, number, person name, and ethnic/national group.

The full per-language tag lists follow:

```
"fr": [  
  "LOC",  
  "MISC",  
  "ORG",  
  "PER"  
],  
"en": [  
  "CARDINAL",  
  "DATE",  
  "EVENT",  
  "FAC",  
  "GPE",  
  "LANGUAGE",  
  "LAW",  
  "LOC",  
  "MONEY",
```

```
"NORP",
"ORDINAL",
"ORG",
"PERCENT",
"PERSON",
"PRODUCT",
"QUANTITY",
"TIME",
"WORK_OF_ART"
],
"zh-hans": [
  "CARDINAL",
  "DATE",
  "EVENT",
  "FAC",
  "GPE",
  "LANGUAGE",
  "LAW",
  "LOC",
  "MONEY",
  "NORP",
  "ORDINAL",
  "ORG",
  "PERCENT",
  "PERSON",
  "PRODUCT",
  "QUANTITY",
  "TIME",
  "WORK_OF_ART"
],
"ru": [
  "LOC",
  "MISC",
  "ORG",
  "PER"
],
"uk": [
  "LOC",
  "MISC",
  "ORG",
  "PERS"
],
"ar": [
  "LOC",
```

```
    "MISC",
    "ORG",
    "PER"
  ],
  "hu": [
    "LOC",
    "MISC",
    "ORG",
    "PER"
  ],
  "af": [
    "LOC",
    "MISC",
    "ORG",
    "PERS"
  ],
  "bg": [
    "EVT",
    "LOC",
    "ORG",
    "PER",
    "PRO"
  ],
  "fi": [
    "DATE",
    "EVENT",
    "LOC",
    "ORG",
    "PER",
    "PRO"
  ],
  "my": [
    "LOC",
    "NE",
    "NUM",
    "ORG",
    "PNAME",
    "RACE",
    "TIME"
  ],
  "it": [
    "LOC",
    "ORG",
    "PER"
```

```
],
"de": [
  "LOC",
  "MISC",
  "ORG",
  "PER"
],
"nl": [
  "LOC",
  "MISC",
  "ORG",
  "PER"
],
"vi": [
  "LOCATION",
  "MISCELLANEOUS",
  "ORGANIZATION",
  "PERSON"
],
"es": [
  "LOC",
  "MISC",
  "ORG",
  "PER"
]
```